Schedule for TQC 2018 and QSML 2018

	Monday, 16.07.2018	Tuesday, 17.07.2018	Wednesday, 18.07.2018	Thursday, 19.07.2018	Friday, 20.07.2018
8:45	Registration				
9:00		Invited: Hakop Pashayan	Talk 17: Yfke Dulek	Opening Remarks	
9:15	Opening Remarks		Talk 18: Andrew Glaudell	Dave Bacon	Nathan Wiebe
9:25	Invited: Joe Fitzsimons		Talk 19: Matthew Amy		
9:50		Talk 9: Angela Karanjai			
10:00				Coffee break	
10:15	Coffee break		Talk 20: Adam Bouland		
10:30				Katja Ried	Mária Kieferová
10:40			Coffee break		
10:45	Talk 1: Eyal Bairey	Talk 10: Seunghoan Song			
11:10	Talk 2: Mario Berta	Talk 11: Kun Fang	Keynote: Michelle Simmons		
11:15				Maria Schuld	Srinivasan Arunachalam
11:35	Talk 3: Angelo Lucia	Talk 12: Ángela Capel			
12:00	Lunch Break			Lunch Break	
12:15			Lunch Break		
14:00	Talk 4: Chinmay Nirkhe	Invited: Adam Bouland	Talk 21: Zoltan Zimboras	Axel Dahlberg	Nana Liu
14:25	Talk 5: Itamar Vigdorovich		Talk 22: Raymond Wong	Lídia Del Rio	Min-Hsiu Hsieh
14:50	Talk 6: Stephen Piddock	Talk 13: Ansis Rosmanis	Talk 23: Pooya Ronagh & Nishad Maskara		
15:00				Coffee break	
15:15	Coffee break				
15:30				Giacomo Torlai	Sally Shrapnel
15:45	Talk 7: Kostyantyn Kechedzhi & Ryan Babbush	Talk 14: Christopher Cade	Talk 24: Christophe Vuillot		
16:00				Justyna P. Zwolak	Virginia Frey
16:10	Talk 8: Konstantyn Kechedzhi	Talk 15: Alexander Helm	Talk 25: Christopher Chubb		
16:30				Quantum Leaps!	
16:35		Talk 16: Simon Perdrix	Talk 26: Paul Webster		
17:00	Poster Session		Industry Session	Poster Session (with light dinner)	
18:00		Conference Dinner			
19:00					
22:00					

#	Talk title	Authors
1	Learning phase transitions from dynamics	Evert van Nieuwenburg, Eyal Bairey and Gil Refael
2	Thermal States as Convex Combinations of Matrix Product States	Mario Berta, Fernando Brandao, Jutho Haegeman, Volkher Scholz and Frank Verstraete
3	Locality at the boundary implies gap in the bulk for 2D PEPS	Michael Kastoryano, Angelo Lucia and David Perez-Garcia
4	Approximate low-weight check codes and circuit lower bounds for noisy ground states	Chinmay Nirkhe, Umesh V. Vazirani and Henry Yuen
5	On the Complexity of Two Dimensional Commuting Local Hamiltonians	Dorit Aharonov, Oded Kenneth and Itamar Vigdorovich
6	Universal Qudit Hamiltonians	Stephen Piddock and Ashley Montanaro
7	Quantum algorithms to simulate many-body physics of correlated fermions	Zhang Jiang, Kevin Sung, Kostyantyn Kechedzhi, Vadim Smelyanskiy and Sergio Boixo
	Quantum Simulation of Electronic Structure with Linear Depth and Connectivity	Ian Kivlichan, Jarrod McClean, Nathan Wiebe, Craig Gidney, Alán Aspuru-Guzik, Garnet Chan and Ryan Babbush
8	Non-ergodic delocalized states for efficient population transfer within a narrow band of the energy landscape	Vadim Smelyanskiy, Konstantyn Kechedzhi, Sergio Boixo, Sergei Isakov, Hartmut Neven and Boris Altshuler
9	Contextuality bounds the efficiency of classical simulation of quantum processes	Angela Karanjai, Joel Wallman and Stephen Bartlett
10	Quantum Network Code for Multiple-Unicast Network with Quantum Invertible Linear Operations	Seunghoan Song and Masahito Hayashi
11	Quantum Channel Simulation and the Channel's Smooth Max-Information	Kun Fang, Xin Wang, Marco Tomamichel and Mario Berta
12	Quantum conditional relative entropy and quasi-factorization of the relative entropy	Angela Capel, Angelo Lucia and David Perez-Garcia
13	Quantum Lower Bounds for Tripartite Versions of the Hidden Shift and the Set Equality Problems	Aleksandrs Belovs and Ansis Rosmanis
14	The Quantum Complexity of Computing Schatten p-norms	Christopher Cade and Ashley Montanaro
15	Subset Sum Quantumly in (1.17)^n	Alexander Helm and Alexander May
16	A Complete Axiomatisation of the ZX-Calculus for Clifford+T Quantum Mechanics	Emmanuel Jeandel, Simon Perdrix and Renaud Vilmart
17	Quantum ciphertext authentication and key recycling with the trap code	Yfke Dulek and Florian Speelman
18	Canonical forms for single-qutrit Clifford+T operators	Andrew Glaudell, Neil Ross and Jacob Taylor
19	On the CNOT-complexity of CNOT-PHASE circuits	Matthew Amy, Parsiad Azimzadeh and Michele Mosca
20	Trading inverses for an irrep in the Solovay-Kitaev theorem	Adam Bouland and Maris Ozols
21	Universal extensions of restricted classes of quantum operations	Michal Oszmaniec and Zoltan Zimboras
22	Two-Qubit Stabilizer Circuits with Recovery I: Existence	Wim van Dam and Raymond Wong
	Two-Qubit Stabilizer Circuits with Recovery II: Analysis	Wim van Dam and Raymond Wong
23	Deep neural decoders for near term fault-tolerant experiments	Christopher Chamberland and Pooya Ronagh
	Advantages of versatile neural-network decoding for topological codes	Nishad Maskara, Aleksander Kubica and Tomas Jochym-O'Connor
24	Quantum Error Correction with the Surface-GKP Code	Christophe Vuillot, Hamad Asasi, Yang Wang, Leonid Pryadko and Barbara Terhal
25	Statistical mechanical models for correlated noise in arbitrary stabiliser codes	Christopher Chubb and Steve Flammia
26	Locality-preserving logical operators in topological stabilizer codes	Paul Webster and Stephen Bartlett

Poster session

#	Poster title	Authors
1	Conditional quantum entropy power inequality for d-level quantum systems	Kabgyun Jeong, Soojoon Lee and Hyunseok Jeong
2	Activation of quantum capacity of Gaussian channels	Youngrong Lim and Soojoon Lee
3	State exchange with quantum side information	Yonghae Lee and Soojoon Lee
4	Tsallis entropy and general polygamy of multiparty quantum entanglement in arbitrary dimensions	Jeong San Kim
5	Analytical criterion for extremal quantum correlations in the simplest Bell scenario	Satoshi Ishizaka
6	Free energy-based reinforcement learning using a quantum processor	Anna Levit, Daniel Crawford, Navid Ghadermarzy, Jaspreet Oberoi, Ehsan Zahedinejad and Pooya Ronagh
7	Bipartite discrimination of independently prepared quantum states as a counterexample of a parallel repetition	Seiseki Akibue and Go Kato
8	Analysing correlated noise on the surface code using adaptive decoding algorithms	Naomi Nickerson and Benjamin Brown
9	Quantum Contextuality Monogamy and Quantum Key Distribution	Arvind, Jaskaran Singh and Kishor Bharti
10	Convex resource theory of non-Gaussianity	Ryuji Takagi and Quntao Zhuang
11	Coherifying quantum channels	Kamil Korzekwa, Stanisław Czachórski, Zbigniew Puchała and Karol Życzkowski
12	Beyond the thermodynamic limit: finite-size corrections to state interconversion rates	Christopher Chubb, Marco Tomamichel and Kamil Korzekwa
13	A Matrix Exponentiated Gradient Algorithm for Online Quantum State Estimation	Akram Youssry, Christopher Ferrie and Marco Tomamichel
14	Bayesian quantum state estimation under Bregman divergence	Maria Quadeer, Christopher Ferrie and Marco Tomamichel
15	Synthesis of Logical Clifford Operators via Symplectic Geometry	Narayanan Rengaswamy, Robert Calderbank, Swanand Kadhe and Henry Pfister
16	Quantum correlations and global coherence in distributed quantum computing	Farid Shahandeh, Austin Lund and Timothy Ralph
17	Measurement Contextuality and Planck's Constant	Lucas Kocia and Peter Love
18	A quantum algorithm for simulating non-sparse Hamiltonians	Chunhao Wang and Leonard Wossnig
19	Nonstationary cavity quantum electrodynamics with semiconductor one-dimensional photonic crystals	Diego Nicolás Bernal-García, Boris A. Rodríguez and Herbert Vinck-Posada
20	Spectra of Gauge Code Hamiltonians	Simon Burton
21	Inadequacy of modal logic in quantum settings	Nuriya Nurgalieva and Lídia Del Rio
22	General framework for multi-resource theories and the first law	Lidia Del Rio, Jonathan Oppenheim and Carlo Sparaciari
23	Universal complex conjugation of unitaries	Jisho Miyazaki, Akihito Soeda and Mio Murao
24	One-shot exact quantum state merging and splitting	Hayata Yamasaki and Mio Murao
25	Improved Techniques for Preparing Eigenstates of Fermionic Hamiltonians	Dominic Berry, Maria Kieferova, Artur Scherer, Yuval Sanders, Guang Low, Nathan Wiebe, Craig Gidney and Ryan Babbush
26	Simulating the dynamics of time-dependent Hamiltonians with a truncated Dyson series	Maria Kieferova, Artur Scherer and Dominic Berry
27	Contextuality as a resource for measurement-based quantum computation beyond qubits	Markus Frembs, Sam Roberts and Stephen Bartlett
28	Practical, reliable error bars for quantum channels	Thinh Le, Philippe Faist, Jonas Helsen, David Elkouss and Stephanie Wehner
29	Wavelet-Based Representations of Quantum Field Theory	Yuval Sanders, Bryte Hagan, Dean Southwood, Sukhwinder Singh, Barry Sanders and Gavin Brennen
30	Linear-time general decoding algorithm for the surface code	Andrew Darmawan and David Poulin
31	Quantum advantage of unitary Clifford circuits with magic state inputs	Mithuna Yoganathan, Richard Jozsa and Sergii Strelchuk
32	Improved key rate for RRDPS quantum key distribution without block-wise phase randomization	Takaya Matsuura, Toshihiko Sasaki and Masato Koashi
33	Mapping NP-Hard Problems to Restricted Adiabatic Quantum Architectures	Gary Mooney, Sam Tonetto, Charles Hill and Lloyd Hollenberg
34	Matrix Product States simulations of noisy IQP circuits	Charles Hill, Michael Bremner and Lloyd Hollenberg
35	Quantum Modeling of Sparse Information for Machine Learning	Syed Adil Rab
36	Extreme points of quantum correlations	Thinh Le, Antonios Varvitsiotis and Yu Cai
37	Majorana dimer models and quantum error correction	Alexander Jahn, Marek Gluza, Fernando Pastawski and Jens Eisert
38	Quantum algorithms for planning problems using quantum walk techniques	Stephen Piddock
39	Towards Large-scale Functional Verification of Universal Quantum Circuits	Matthew Amy
40	Learning Hamiltonians from local measurements	Eyal Bairey, Netanel Lindner and Itai Arad